up to B where the splitting occurred. The extra path travelled by the wave transmitted from D is the path inside the film, $BC + CD$. If we approximate the incidence to be nearly normal ($i = 0$) and the flim of small thickness, then the points B and D are very close to each other. The extra distance travelled by the wave is approximately twice thickness of the film, $BC + CD = 2d$. As this extra path is traversed inside the medium of refractive index μ, the optical path difference is, $\delta = 2\mu d$.

The condition for constructive interference in transmitted ray is,

$$2\mu d = n\lambda \qquad (7.33)$$

Similarly, the condition for destructive interference in transmitted ray is,

$$2\mu d = (2n-1)\frac{\lambda}{2} \qquad (7.34)$$

For reflected light

It is experimentally and theoretically proved that a wave while travelling in a rarer medium and getting reflected by a denser medium, undergoes a phase change of π. Hence, an additional path difference of $\lambda/2$ should be considered for reflected light.

Let us consider the path difference between the light reflected by the upper surface at A and the other coming out at C after passing through the film. The additional path travelled by the light coming out from C is the path inside the film, $AB + BC$. For near normal incidence and film of small thickness, this distance could be approximated as, $AB + BC = 2d$. As this extra path is travelled in the medium of refractive index μ, the optical path difference is, $\delta = 2\mu d$.

The condition for constructive interference for reflected ray is,

$$2\mu d + \frac{\lambda}{2} = n\lambda \text{ (or) } 2\mu d = (2n-1)\frac{\lambda}{2} \qquad (7.35)$$

The additional path difference $\lambda/2$ is due to the phase change of π in rarer to denser reflection taking place at A.

The condition for destructive interference for reflected ray is,

$$2\mu d + \frac{\lambda}{2} = (2n+1)\frac{\lambda}{2} \text{ (or) } 2\mu d = n\lambda \qquad (7.36)$$

Note

If the incidence is not nearly normal but at an angle of incidence i which has an angle of refraction r, then the expression $2\mu d$ is to be replaced with $2\mu d \cos r$.

EXAMPLE 7.8

Find the minimum thickness of a film of refractive index 1.25, which will strongly reflect the light of wavelength 589 nm. Also find the minimum thickness of the film to be anti-reflecting.

Solution

λ = 589 nm = 589×10^{-9} m

For the film to have strong reflection, the reflected waves should interfere constructively. The least optical path difference introduced by the film should be $\lambda/2$. The optical path difference between the waves reflected from the two surfaces of the film is $2\mu d$. Thus, for strong reflection, $2\mu d = \lambda/2$ [As given in equation (7.35). with $n = 1$]

Rewriting, $d = \dfrac{\lambda}{4\mu}$

Substituting, $d = \dfrac{589\times10^{9}}{4\times1.25} = 117.8\times10^{-9}$

$d = 117.8\times10^{-9} = 117.8$ nm

XII_Physics_Vol-2_UNIT-7_EM.indd 68 03-03-2022 15:18:56